

The four Hall–Littlewood partial-symmetrizer identities

(L1)–(L4)

A closed form for the level-one Bernstein symmetrizer on $x_1^a e_k(\widehat{x}_1)$

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Abstract

Rick (Day 204, §4a) reduced his Sub-Lemma Z to four q -free Hall–Littlewood partial-symmetrizer identities (L1)–(L4), verified numerically at $r = 2, 3, 4, 5$ and graded **sketched**, with an explicit handoff: “natural attack via Macdonald III.5 Pieri for HL polynomials or Ram–Yip alcove-walks.” We prove all four, for every $r \geq 1$ and every number of variables $m \geq 1$ simultaneously. Neither Pieri nor alcove walks are needed. The four identities are corollaries of a single closed form (Theorem 2) for the level-one Bernstein symmetrizer $\sigma_m = \sum_{k=0}^{m-1} T_k T_{k-1} \cdots T_1$ applied to $x_1^a e_k(x_2, \dots, x_m)$, whose coefficients are the one-row Hall–Littlewood polynomials $P_{(n)}(x; t)$. The proof is elementary and self-contained: an induction on m evaluating σ_m on the powers x_1^a , followed by Λ_m -linearity.

1 Statement

1.1 Setup and conventions

Fix $m \geq 1$ and let $x = (x_1, \dots, x_m)$. Write $\Lambda_m = \mathbb{Q}(t)[x]^{S_m}$ for the ring of symmetric polynomials, $e_j = e_j(x_1, \dots, x_m)$ for the elementary symmetric polynomials (with $e_0 = 1$ and $e_j = 0$ for $j < 0$ or $j > m$), and $E(u) = \sum_{j \geq 0} e_j u^j = \prod_{j=1}^m (1 + x_j u)$. We write $\widehat{x}_i$ for the variable list with x_i deleted, so $e_k(\widehat{x}_1) = e_k(x_2, \dots, x_m)$ — the *tail*. Set $[n]_t = 1 + t + \cdots + t^{n-1}$ for $n \geq 1$ and $[n]_t = 0$ for $n \leq 0$.

The Demazure–Lusztig operators act on $\mathbb{Q}(t)[x]$ by

$$T_i f = t f + \frac{t x_i - x_{i+1}}{x_i - x_{i+1}} (s_i f - f), \quad 1 \leq i \leq m-1, \quad (1)$$

where s_i transposes x_i, x_{i+1} . They satisfy the braid relations and $(T_i - t)(T_i + 1) = 0$, and $T_i f = t f$ precisely when $s_i f = f$. The *level-one Bernstein partial symmetrizer* is

$$\sigma_m = \sum_{k=0}^{m-1} T_k T_{k-1} \cdots T_1 = 1 + T_1 + T_2 T_1 + \cdots + T_{m-1} T_{m-2} \cdots T_1, \quad (2)$$

the $k = 0$ summand being the identity. Equivalently $\sigma_m = \sum_{k=0}^{m-1} T_{w_k}$ where $w_k = s_k s_{k-1} \cdots s_1$ runs over the minimal-length representatives of $S_m / S_{\{2, \dots, m\}}$; consequently $\sum_{w \in S_m} T_w = \sigma_m \cdot \sum_{v \in S_{\{2, \dots, m\}}} T_v$, which is why σ_m carries tail-symmetric polynomials to symmetric ones.

Finally, let $q_n = q_n(x; t)$ be the Hall–Littlewood q -functions, defined by

$$Q(u) = \sum_{n \geq 0} q_n u^n = \prod_{j=1}^m \frac{1 - tx_j u}{1 - x_j u}, \quad (3)$$

and put $\tilde{q}_n := q_n/(1 - t)$ for $n \geq 1$, $\tilde{q}_0 := 1$.

Lemma 1. *q_n is divisible by $1 - t$ in Λ_m for every $n \geq 1$; hence $\tilde{q}_n \in \Lambda_m$. Moreover $\tilde{q}_n = P_{(n)}(x; t)$, the one-row Hall–Littlewood P -polynomial, and*

$$\tilde{q}_1 = e_1, \quad \tilde{q}_2 = e_1^2 - [2]_t e_2.$$

Proof. From (3), $q_n = \sum_{k+l=n} (-t)^k e_k h_l$ where h_l are the complete homogeneous symmetric polynomials. At $t = 1$ this is $\sum_k (-1)^k e_k h_{n-k}$, which vanishes for $n \geq 1$ by the Newton relation $\sum_{k \geq 0} (-1)^k e_k h_{n-k} = 0$ ($n \geq 1$). Hence $(1 - t) \mid q_n$. The identification $q_n = (1 - t)P_{(n)}$ is Macdonald III (2.8)–(2.10); it is not used below. The two displayed values follow from $q_1 = h_1 - te_1 = (1 - t)e_1$ and $q_2 = h_2 - te_1 h_1 + t^2 e_2 = (1 - t)e_1^2 - (1 - t^2)e_2$ using $h_2 = e_1^2 - e_2$. $\square$

1.2 The results

Theorem 2 (Master Lemma). *For all $m \geq 1$, all $a \geq 1$ and all $k \geq 0$,*

$$\sigma_m[x_1^a e_k(\hat{x}_1)] = M(a, k) := \sum_{l=0}^k (-1)^l e_{k-l} \tilde{q}_{a+l} = (-1)^{a+1} \left([k+a]_t e_{k+a} + \sum_{n=1}^{a-1} (-1)^n \tilde{q}_n e_{k+a-n} \right). \quad (4)$$

In particular

$$M(1, k) = [k+1]_t e_{k+1}, \quad (5)$$

$$M(2, k) = -[k+2]_t e_{k+2} + e_1 e_{k+1}, \quad (6)$$

$$M(3, k) = [k+3]_t e_{k+3} - e_1 e_{k+2} + (e_1^2 - [2]_t e_2) e_{k+1}. \quad (7)$$

Theorem 3 ((L1)–(L4); Rick, Day 204 §4a). *For all $m \geq 1$ and all $r \geq 1$, with $e_\lambda := \prod_i e_{\lambda_i}$,*

$$(L1) \quad \sigma_m[x_1 e_r(\hat{x}_1) e_1(\hat{x}_1)] = [r+2]_t e_{r+2} + t[r]_t e_{(r+1,1)}, \quad (8)$$

$$(L2) \quad \sigma_m[x_1^2 e_r(\hat{x}_1)] = -[r+2]_t e_{r+2} + e_{(r+1,1)}, \quad (9)$$

$$(L3) \quad \sigma_m[x_1^2 e_{r-1}(\hat{x}_1) e_1(\hat{x}_1)] = -[r+2]_t e_{r+2} - t[r]_t e_{(r+1,1)} + [2]_t e_{(r,2)}, \quad (10)$$

$$(L4) \quad \sigma_m[x_1^3 e_{r-1}(\hat{x}_1)] = [r+2]_t e_{r+2} - e_{(r+1,1)} - [2]_t e_{(r,2)} + e_{(r,1,1)}. \quad (11)$$

Remark 4. The identities are uniform in m : no hypothesis $m \geq r + 2$ is needed. For small m they remain true with the convention $e_j = 0$ for $j > m$, both sides then degenerating consistently. They are also uniform in r : there is no exceptional small- r case. This is worth stating because the obvious route — substituting $u := t^r$ to turn the t -integers $[r]_t, [r+2]_t$ into rational functions — does *not* carry the convention $[k]_t = 0$ for $k \leq 0$, and small r would have to be checked separately. The proof below never substitutes for t^r ; the r -dependence is carried entirely by the index k of e_k , and (4) is an identity of polynomials in t valid for every integer $k \geq 0$.

2 Proof of the Master Lemma

The proof has three steps: a one-variable computation (Lemma 6), an induction on m that evaluates σ_m on the powers x_1^a (Proposition 8), and Λ_m -linearity (Lemma 5).

Lemma 5. *For $g \in \Lambda_m$ and any f , $T_i(gf) = gT_if$ for all i ; hence $\sigma_m(gf) = g\sigma_m(f)$.*

Proof. g is in particular symmetric in x_i, x_{i+1} , so $s_i(gf) = g s_i f$, and (1) gives $T_i(gf) = tgf + \frac{tx_i - x_{i+1}}{x_i - x_{i+1}} g(s_i f - f) = gT_if$. The claim for σ_m follows by iterating and summing. $\square$

Lemma 6. *For $c \geq 1$,*

$$T_i(x_i^c) = x_{i+1}^c + (1-t) \sum_{p=1}^{c-1} x_i^p x_{i+1}^{c-p}.$$

Proof. Write $A = x_i$, $B = x_{i+1}$. Since $s_i A^c = B^c$ and $\frac{B^c - A^c}{A - B} = -\sum_{p=0}^{c-1} A^p B^{c-1-p}$, (1) gives

$$T_i A^c = tA^c - (tA - B) \sum_{p=0}^{c-1} A^p B^{c-1-p} = tA^c - t \sum_{p=1}^c A^p B^{c-p} + \sum_{p=0}^{c-1} A^p B^{c-p}.$$

The term tA^c cancels against the $p = c$ term of the middle sum, and the $p = 0$ term of the last sum is B^c . What remains is $B^c + (1-t) \sum_{p=1}^{c-1} A^p B^{c-p}$. $\square$

Write $\tilde{q}_b^{(j)} := \tilde{q}_b(x_1, \dots, x_j; t)$ for the truncation to the first j variables, with $\tilde{q}_0^{(j)} = 1$ and $\tilde{q}_b^{(0)} = 0$ for $b \geq 1$.

Lemma 7. *For $j \geq 1$ and $c \geq 1$,*

$$\tilde{q}_c^{(j)} - \tilde{q}_c^{(j-1)} = x_j^c + (1-t) \sum_{b=1}^{c-1} \tilde{q}_b^{(j-1)} x_j^{c-b}.$$

Proof. By (3), $Q^{(j)}(u) = Q^{(j-1)}(u) \cdot \frac{1-tx_j u}{1-x_j u}$, hence

$$Q^{(j)}(u) - Q^{(j-1)}(u) = Q^{(j-1)}(u) \frac{(1-t)x_j u}{1-x_j u} = (1-t) Q^{(j-1)}(u) \sum_{c \geq 1} x_j^c u^c.$$

Taking the coefficient of u^c ($c \geq 1$) gives $q_c^{(j)} - q_c^{(j-1)} = (1-t) \sum_{b=0}^{c-1} q_b^{(j-1)} x_j^{c-b}$. Divide by $1-t$ and split off the $b = 0$ term, using $q_0 = 1$ and $q_b = (1-t)\tilde{q}_b$ for $b \geq 1$. $\square$

Proposition 8. *Set $\rho_j := T_{j-1}T_{j-2} \cdots T_1$ (so $\rho_1 = \text{id}$), so that $\sigma_m = \sum_{j=1}^m \rho_j$. Then for every $a \geq 1$ and every $1 \leq j \leq m$,*

$$\rho_j(x_1^a) = x_j^a + (1-t) \sum_{b=1}^{a-1} \tilde{q}_b^{(j-1)} x_j^{a-b}, \quad (12)$$

and consequently

$$\sigma_m(x_1^a) = \tilde{q}_a(x_1, \dots, x_m; t) \quad (a \geq 1), \quad \sigma_m(1) = [m]_t. \quad (13)$$

Proof. We prove (12) by induction on j .

Base $j = 1$. $\rho_1(x_1^a) = x_1^a$, and the sum on the right is empty because $\tilde{q}_b^{(0)} = 0$ for $b \geq 1$.

Step. Assume (12) for $j - 1$, i.e. $\rho_{j-1}(x_1^a) = x_{j-1}^a + (1 - t) \sum_{b=1}^{a-1} \tilde{q}_b^{(j-2)} x_{j-1}^{a-b}$. Apply $\rho_j = T_{j-1} \rho_{j-1}$. Each $\tilde{q}_b^{(j-2)}$ is a polynomial in $x_1, \dots, x_{j-2}$ alone, hence is fixed by T_{j-1} by Lemma 5; and every exponent $a - b$ occurring satisfies $a - b \geq 1$, so Lemma 6 applies to each power of x_{j-1} . Thus

$$\rho_j(x_1^a) = \left[x_j^a + (1 - t) \sum_{p=1}^{a-1} x_{j-1}^p x_j^{a-p} \right] + (1 - t) \sum_{b=1}^{a-1} \tilde{q}_b^{(j-2)} \left[x_j^{a-b} + (1 - t) \sum_{p=1}^{a-b-1} x_{j-1}^p x_j^{a-b-p} \right].$$

Collect the coefficient of x_j^{a-c} for each c with $1 \leq c \leq a - 1$. It receives x_{j-1}^c from the first bracket ($p = c$), $\tilde{q}_c^{(j-2)}$ from the second bracket's first term ($b = c$), and $(1 - t) \sum_{b=1}^{c-1} \tilde{q}_b^{(j-2)} x_{j-1}^{c-b}$ from the double sum ($b + p = c$, $b, p \geq 1$); each of these carries the overall prefactor $(1 - t)$ except the first bracket's, which carries it too. Hence the coefficient of $(1 - t)x_j^{a-c}$ is

$$x_{j-1}^c + \tilde{q}_c^{(j-2)} + (1 - t) \sum_{b=1}^{c-1} \tilde{q}_b^{(j-2)} x_{j-1}^{c-b} = \tilde{q}_c^{(j-1)}$$

by Lemma 7 applied with $j - 1$ in place of j . The coefficient of x_j^a is 1. This is exactly (12) for j .

Summing (12) over $j = 1, \dots, m$ and using Lemma 7 once more telescopes:

$$\sigma_m(x_1^a) = \sum_{j=1}^m \left(x_j^a + (1 - t) \sum_{b=1}^{a-1} \tilde{q}_b^{(j-1)} x_j^{a-b} \right) = \sum_{j=1}^m (\tilde{q}_a^{(j)} - \tilde{q}_a^{(j-1)}) = \tilde{q}_a^{(m)} - \tilde{q}_a^{(0)} = \tilde{q}_a.$$

Finally $T_i 1 = t$, so $\rho_j(1) = t^{j-1}$ and $\sigma_m(1) = \sum_{j=1}^m t^{j-1} = [m]_t$. $\square$

Proof of Theorem 2. The polynomial identity $e_k(\hat{x}_1) = \sum_{l=0}^k (-x_1)^l e_{k-l}$ (equivalently $\prod_{j \geq 2} (1 + x_j u) = E(u)/(1 + x_1 u)$) gives

$$x_1^a e_k(\hat{x}_1) = \sum_{l=0}^k (-1)^l e_{k-l} x_1^{a+l}.$$

Each e_{k-l} lies in Λ_m , so Lemma 5 and Proposition 8 give, for $a \geq 1$ (so that every exponent $a + l \geq 1$),

$$\sigma_m[x_1^a e_k(\hat{x}_1)] = \sum_{l=0}^k (-1)^l e_{k-l} \sigma_m(x_1^{a+l}) = \sum_{l=0}^k (-1)^l e_{k-l} \tilde{q}_{a+l},$$

which is the first equality in (4).

For the closed form, multiply by u^k and sum over $k \geq 0$:

$$\sum_{k \geq 0} M(a, k) u^k = \sum_{l \geq 0} (-u)^l \tilde{q}_{a+l} \sum_{k \geq l} e_{k-l} u^{k-l} = E(u) \sum_{l \geq 0} (-u)^l \tilde{q}_{a+l}.$$

Now $\sum_{l \geq 0} (-u)^l q_{a+l} = (-u)^{-a} [Q(-u) - \sum_{n=0}^{a-1} q_n (-u)^n]$, and by (3) $Q(-u) = \prod_j \frac{1+tx_j u}{1+x_j u} = E(tu)/E(u)$. Therefore

$$\sum_{k \geq 0} M(a, k) u^k = \frac{(-1)^a}{(1-t) u^a} \left[E(tu) - E(u) \sum_{n=0}^{a-1} (-1)^n q_n u^n \right].$$

Extracting the coefficient of u^k , i.e. the coefficient of u^{k+a} inside the bracket, and using $[u^{k+a}]E(tu) = t^{k+a} e_{k+a}$, $q_0 = 1$:

$$M(a, k) = \frac{(-1)^a}{1-t} \left[(t^{k+a} - 1) e_{k+a} - \sum_{n=1}^{a-1} (-1)^n q_n e_{k+a-n} \right] = (-1)^{a+1} \left[[k+a]_t e_{k+a} + \sum_{n=1}^{a-1} (-1)^n \tilde{q}_n e_{k+a-n} \right],$$

using $\frac{t^N - 1}{1-t} = -[N]_t$ and $\tilde{q}_n = q_n/(1-t)$. Specialising $a = 1, 2, 3$ and inserting $\tilde{q}_1 = e_1$, $\tilde{q}_2 = e_1^2 - [2]_t e_2$ (Lemma 1) gives (5)–(7). $\square$

3 Proof of (L1)–(L4)

Lemma 9 (Product Lemma). *For $a \geq 1$ and $k \geq 0$,*

$$\sigma_m [x_1^a e_k(\hat{x}_1) e_1(\hat{x}_1)] = e_1 M(a, k) - M(a+1, k).$$

Proof. $e_1(\hat{x}_1) = e_1 - x_1$, so $x_1^a e_k(\hat{x}_1) e_1(\hat{x}_1) = e_1 \cdot x_1^a e_k(\hat{x}_1) - x_1^{a+1} e_k(\hat{x}_1)$. Apply σ_m , using Lemma 5 for the symmetric factor e_1 and Theorem 2 for each term. $\square$

Proof of Theorem 3. Throughout, $[r+1]_t - 1 = t + t^2 + \cdots + t^r = t[r]_t$.

(L2). By (6) with $k = r$,

$$\sigma_m [x_1^2 e_r(\hat{x}_1)] = M(2, r) = -[r+2]_t e_{r+2} + e_1 e_{r+1} = -[r+2]_t e_{r+2} + e_{(r+1,1)}.$$

(L4). By (7) with $k = r-1$,

$$\sigma_m [x_1^3 e_{r-1}(\hat{x}_1)] = M(3, r-1) = [r+2]_t e_{r+2} - e_1 e_{r+1} + (e_1^2 - [2]_t e_2) e_r,$$

which is $[r+2]_t e_{r+2} - e_{(r+1,1)} - [2]_t e_{(r,2)} + e_{(r+1,1)}$.

(L1). By Lemma 9 with $a = 1, k = r$, then (5), (6):

$$\sigma_m [x_1 e_r(\hat{x}_1) e_1(\hat{x}_1)] = e_1 M(1, r) - M(2, r) = [r+1]_t e_1 e_{r+1} + [r+2]_t e_{r+2} - e_1 e_{r+1},$$

i.e. $[r+2]_t e_{r+2} + ([r+1]_t - 1) e_{(r+1,1)} = [r+2]_t e_{r+2} + t[r]_t e_{(r+1,1)}$.

(L3). By Lemma 9 with $a = 2, k = r-1$, then (6), (7):

$$\begin{aligned} \sigma_m [x_1^2 e_{r-1}(\hat{x}_1) e_1(\hat{x}_1)] &= e_1 M(2, r-1) - M(3, r-1) \\ &= e_1 (-[r+1]_t e_{r+1} + e_1 e_r) - ([r+2]_t e_{r+2} - e_1 e_{r+1} + (e_1^2 - [2]_t e_2) e_r) \\ &= -[r+2]_t e_{r+2} + (1 - [r+1]_t) e_1 e_{r+1} + (e_1^2 - e_1^2 + [2]_t e_2) e_r \\ &= -[r+2]_t e_{r+2} - t[r]_t e_{(r+1,1)} + [2]_t e_{(r,2)}. \end{aligned}$$

The two $e_1^2 e_r$ terms cancel; this cancellation is the whole content of (L3), and it is exactly the step a naive index-shift gets wrong (see §4). $\square$

Corollary 10. *Rick's residual gap in the reduction of Sub-Lemma Z (Day 204, §4a) is closed. The four identities hold for all $r \geq 1$ and all $m \geq 1$, and hence so does the assembly $\sigma_m \cdot \pi(e_r \cdot e_1)$ with the q -weights $\{1, q^{-1}, q^{-1}, q^{-2}\}$.*

4 Where a shorter route breaks

The first route I took was a residue computation. Writing $F(y) = \prod_j \frac{y-tx_j}{y-x_j}$ and using $\prod_{j \neq i} \frac{x_i-tx_j}{x_i-x_j} = \frac{1}{x_i(1-t)}$ $\text{Res}_{y=x_i} F$, one gets from the transposition formula (Remark 11) that for $a \geq 1$

$$\sigma_m[x_1^a \Phi(\widehat{x}_1)] = \frac{1}{1-t} \sum_{l \geq 0} \varphi_l q_{a+l}, \quad \text{where } \Phi(\widehat{x}_i) = \sum_l \varphi_l x_i^l,$$

because the sum of all residues of $y^{a-1} \Phi^\sharp(y) F(y)$ on $\mathbb{P}^1$ vanishes and the residue at ∞ picks out $\sum_l \varphi_l q_{a+l}$ from $F(y) = \sum_n q_n y^{-n}$. This is correct. The error was in expanding $\Phi = e_k e_1$: from

$$e_k(\widehat{x}_i) e_1(\widehat{x}_i) = \left(\sum_{l \geq 0} (-x_i)^l e_{k-l} \right) (e_1 - x_i) = \sum_{L \geq 0} (-x_i)^L e_{k-L} e_1 + \sum_{L \geq 1} (-x_i)^L e_{k-L+1},$$

I read off $\varphi_L = (-1)^L (e_{k-L} e_1 + e_{k-L+1})$ for all $L \geq 0$. But the second sum arises from the substitution $L = l + 1$ and therefore **starts at** $L = 1$: the correct reading is $\varphi_0 = e_k e_1$ and $\varphi_L = (-1)^L (e_{k-L} e_1 + e_{k-L+1})$ only for $L \geq 1$. The spurious $L = 0$ term contributes $e_{k+1} q_a / (1-t)$, which is exactly the difference between $-M(a+1, k)$ and $+e_{k+1} \tilde{q}_a - M(a+1, k)$. Propagated, it turned (L1)'s coefficient $t[r]_t$ into $[r+1]_t$ and (L3)'s term $[2]_t e_{(r,2)}$ into $e_{(r,1,1)}$ — that is, it produced a “correction” to two of Rick’s four identities that were in fact correct as he stated them.

Two things caught it. First, an *internal consistency test*: the erroneous formulas made (L3)–(L2) equal to $e_1 M(2, r-1)$, which combined with (L2) at $r-1$ forces $[2]_t e_2 = e_1^2$ — false. Second, and decisively, the brute-force Hecke computation with (L2) and (L4) used as an *untuned positive control*: Rick had published those two, my derivation reproduced them exactly, so when the same instrument then contradicted my (L1) and (L3) the fault was located in me, not in the instrument or in him.

Lemma 9 exists precisely to remove this hazard. It replaces the index-shifted expansion of $e_k(\widehat{x}_1) e_1(\widehat{x}_1)$ by the identity $e_1(\widehat{x}_1) = e_1 - x_1$ together with Λ_m -linearity, so no re-indexing of a generating-function sum is ever performed.

Remark 11 (The transposition formula). For f symmetric in $x_2, \dots, x_m$ one has

$$\sigma_m f = \sum_{i=1}^m (s_{1i} f) \prod_{j \neq i} \frac{x_i - tx_j}{x_i - x_j},$$

which follows from $\sum_{w \in S_m} T_w f = \sum_{w \in S_m} w(f \prod_{i < j} \frac{x_i - tx_j}{x_i - x_j})$ together with $\sum_{w \in S_m} T_w = \sigma_m \sum_{v \in S_{\{2, \dots, m\}}} T_v$ and $\sum_v v(\prod_{2 \leq i < j} \frac{x_i - tx_j}{x_i - x_j}) = [m-1]_t!$. It is the conceptual source of Proposition 8 — taking $f = x_1^a$ and summing residues recovers $\sigma_m(x_1^a) = \tilde{q}_a$ — but it is *not* used in the proof above, which is why no appeal to the classical Hecke-symmetrizer identity is needed. It is recorded here because it was verified directly against the brute-force Hecke computation at $m \leq 5$ and is the cleanest statement of what σ_m does.

5 Verification

Every statement above was checked symbolically over $\mathbb{Q}(t)$ against an independent brute-force implementation of (1) and (2). The checker uses no formula from the proof: it composes the operators T_i literally (by exact polynomial division), and computes $\tilde{q}_n$ from $q_n = \sum_{k+l=n} (-t)^k e_k h_l$ with h_l from the Newton recursion — not from any of (4), (5)–(7).

Statement	Range checked	Result
Lemma 6	$m \in \{3, 4\}$, all i , $c \leq 5$	30/30
Prop. 8, eq. (12)	$m \leq 5$, $1 \leq j \leq m$, $a \leq 4$	56/56
Prop. 8, eq. (13)	$2 \leq m \leq 6$, $0 \leq a \leq 5$	30/30
Lemma 5	$m \in \{3, 4, 5\}$, $g = e_1, e_2$	6/6
Theorem 2, first equality	$m \in \{3, 4, 5\}$, $1 \leq a \leq 4$, $0 \leq k \leq 3$	48/48
Theorem 2, (5)–(7)	$m \in \{3, 4, 5\}$, $a \leq 3$, $k \leq 4$	45/45
Lemma 9	$m \in \{3, 4, 5\}$, $a \in \{1, 2\}$, $k \leq 3$	24/24
Theorem 3	(m, r) below	36/36
Total: 270 checks, 0 failures.		

For Theorem 3 the pairs are $m = 3$, $r \in \{1, 2\}$; $m = 4$, $r \in \{1, 2, 3\}$; $m = 5$, $r \in \{1, 2, 3, 4\}$. Three of these — $(3, 2)$, $(4, 3)$, $(5, 4)$ — lie in the degenerate range $m < r + 2$, where $e_{r+2} = 0$ and both sides collapse; and every block includes $r = 1$, where $e_{r-1} = e_0 = 1$. Rick’s numerical check covered $r = 2, \dots, 5$ only, at a single m large enough to be generic, so these two edges are new.

The transposition formula of Remark 11 was checked separately, against the same brute-force σ_m , on all four (L)-inputs at $m = 4$ and on x_1^a , $a \leq 3$, at $m = 2$.

Scripts: `proofs/scripts/day2026-09-18/verify_fast.py` (the 270 checks above) and `proofs/scripts/day2026-09-18/verify_fast_output.txt` (the first instrument, which uses rational-function arithmetic and carries the Remark 11 check).

Log: `proofs/scripts/day2026-09-18/verify_fast_output.txt`.

6 What this says about the handoff

Rick’s handoff suggested Macdonald III.5 (Pieri for Hall–Littlewood polynomials) or Ram–Yip alcove walks. Neither is needed, and in hindsight neither is natural: the right-hand sides look like a Pieri expansion — the four partitions of $r + 2$ obtained from (r) by adding two boxes in at most three rows — but that shape is an artifact of the *left*-hand sides, which are x_1^a times a tail-symmetric function of degree $r + 2 - a$ for $a = 1, 2, 3$. The genuine content is a single closed form, Theorem 2, in which the Hall–Littlewood input enters only as the one-row family $\tilde{q}_n = P_{(n)}$ — and only through $n \leq a - 1 \leq 2$. The whole of (L1)–(L4) uses just $\tilde{q}_1 = e_1$ and $\tilde{q}_2 = e_1^2 - [2]_t e_2$.

The operative fact is Proposition 8: σ_m sends $x_1^a \mapsto P_{(a)}$. Everything else is Λ_m -linearity. Since $\{x_1^a\}_{a \geq 0}$ generates the ring of $x_2, \dots, x_m$ -symmetric polynomials as a Λ_m -module, Proposition 8 determines σ_m completely on the domain where it is being used; any further identity of this family is now a finite computation with no new input.